B.3 Pattern inference and execution experiment

The primary objective of this experiment is to disentangle the two stages of reasoning in CoT: pattern inference from in-context demonstrations and pattern execution on test inputs. The prompt instructions for both experiments are provided below:

Prompt Templates

Pattern Inference

```
<regular task description>
<in-context demonstrations>

Now, please perform reasoning to infer the underlying pattern (python function) mapping inputs and outputs.

Your output should strictly follow the json dict format below:
{
    "reasoning": "your reasoning steps",
    "pattern": "your pattern (function)"
}
```

Pattern Execution

```
<regular task description>
<test input>
<ground-truth pattern>

Now, please perform reasoning to apply the above ground-truth pattern to transform the input into output.

Your output should strictly follow the json dict format below:
{
    "reasoning": "your reasoning steps",
    "output": "your output"
}
```

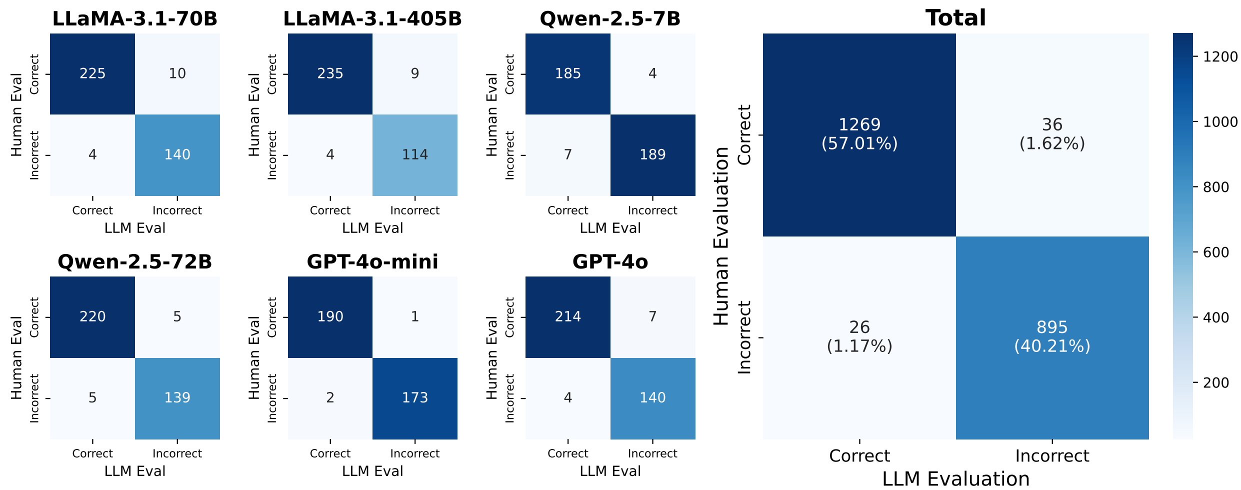

Figure 6: Alignment of human evaluation and LLM evaluation on inferred patterns.

The evaluation of inferred patterns in List Function and MiniSCAN is conducted using different approaches. For List Function, we directly execute Python functions generated by LLMs on all available input data and compare the program outputs with the corresponding ground-truth outputs. Over 95% of the LLM-generated programs successfully compile. For MiniSCAN, the generated rules are expressed as textual descriptions, which makes automated programmatic evaluation challenging. Therefore, we employ Qwen-2.5-72B to assess the correctness of the inferred rules. To evaluate the robustness of this LLM-based assessment, we also conducted human evaluation of rationales, demonstrating strong alignment between the results of LLM evaluation and human evaluation (97.22% total agreement, as shown in Figure 6). Consequently, our evaluation of inferred patterns is deemed reliable.

Below shows an example of the LLM pattern evaluation prompt for Qwen-2.5-72b:

Prompt Templates

Pattern Evaluation

```
You are tasked with judging a sequence-to-sequence problem.

A person is given a series of input and output sequences,
and aims to deduce the rules or word mappings that connect them.

In this scenario, each word in the input sequence can either:

1. Map directly to a word in the output sequence (word mapping).

2. Represent a rule for constructing the output sequence.

Possible Rules for Constructing the Output Sequence
Repeat the former part three times:
Example: If the input is "tmp thri" and "thri" represents this rule, the output should be "tmp tmp tmp."

Swap the former with the latter:
Example: If the input is "tmp1 sw tmp2" and "sw" represents this rule, the output should be "tmp2 tmp1."

Place the latter one between two instances of the former:
Example: If the input is "tmp1 pd tmp2" and "pd" represents this rule, the output should be "tmp1 tmp2 tmp1."

Your Task

You will be provided with the rules or word mappings that the person deduced.
Your objective is to evaluate whether the person correctly deduced these rules or mappings.
If a deduced rule only indicates that a word corresponds to a mapping or construction rule
without specifying what the rule is, it should be deemed incorrect.
```

B.4 Decomposition of explicit and implicit reasoning

The inconsistency between overall CoT performance and the combined performance of explicit pattern inference and execution suggests that an implicit reasoning mechanism may also exist in the latent space of LLMs, despite the use of CoT. In Figure 5, we investigate, across all questions where CoT succeeds, how many questions the LLM fails to infer the correct pattern, while purely implicit reasoning (direct answering) succeeds, as well as the opposite case. These numbers provide intuitive yet practical estimates of the proportions of questions to which explicit and implicit reasoning contribute in CoT's success—proportions that are infeasible to evaluate at scale. For better understanding of this hybrid mechanism, we present a case study[4] for both datasets where CoT succeeds despite incorrect reasoning and inferred patterns. These serve as strong evidence of the contribution of latent reasoning in CoT to pattern-based ICL.

[4]For List Function, we required LLM to generate the python function between CoT reasoning steps and final answer output. In our regular evaluation, python functions are not explicitly required.